

1. Administrative & Study Identification

Full Study Title: Phase III Study of Trastuzumab Deruxtecan (T-DXd) With or Without Pertuzumab Versus Taxane, Trastuzumab and Pertuzumab in HER2-positive, First-line Metastatic Breast Cancer (DESTINY-Breast09) **Short Title/Acronym:** DESTINY-Breast09 **Protocol Number:** D967UC00001 **NCT Number:** NCT04784715 **Posting ID:** 272915588513506993 **Sponsor / Company Name:** AstraZeneca **Investigational Product:** Trastuzumab Deruxtecan (T-DXd / Enhertu) and Pertuzumab (Perjeta) **Phase of Development:** PHASE3 (Phase ID: PHASE3) **Trial Status:** ACTIVE_NOT_RECRUITING (Status Name: ACTIVE_NOT_RECRUITING | Status Description: Active but not currently recruiting) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety—specifically Progression-Free Survival (PFS) by Blinded Independent Central Review (BICR)—of trastuzumab deruxtecan (T-DXd) alone or in combination with pertuzumab, compared to the standard of care (taxane [docetaxel or paclitaxel], trastuzumab, and pertuzumab [THP]) as a first-line treatment in the metastatic setting. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Progression Free Survival (PFS) by Investigator assessment, Duration of Response (DoR), and safety/tolerability profiles, as well as to assess how the treatment and cancer affect patients' quality of life.

2.2 Background & Rationale To address the unmet need for improved initial therapies in HER2-positive advanced or metastatic breast cancer. While T-DXd has demonstrated superior efficacy in previously treated settings, the DESTINY-Breast09 trial evaluates its potential to become the new first-line standard of care by comparing it directly to the established THP regimen, potentially allowing patients to live longer without the cancer getting worse or simply to live longer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (THP: Taxane [docetaxel or paclitaxel], Trastuzumab, and Pertuzumab) **Blinding:** Double-blind (for the T-DXd + placebo arm) and Open-label (for the THP arm) **Randomization:** 1:1:1 across three treatment arms **Phase**

Requirement: Trial must be in PHASE3 (Criteria Type: PHASE_REQUIREMENT)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$):
1,157 adult patients **Number of Sites:** Global trial across approximately 283 locations **Estimated Study Start:** 26-Apr-2021
Estimated Primary Completion: 06-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Key Inclusion Criteria 1. Patients must be ≥ 18 years of age. 2. Pathologically documented breast cancer that is advanced or metastatic. 3. Disease is locally assessed and prospectively centrally confirmed as HER2-positive (IHC 3+ or ISH+). 4. Documented by local testing as hormone receptor (HR)-positive or HR-negative disease in the metastatic setting. 5. No prior chemotherapy or HER2-targeted therapy for advanced or metastatic breast cancer, or only 1 previous line of endocrine therapy in the metastatic setting. (Participants who have received chemotherapy or HER2-targeted therapy in the neo-adjuvant or adjuvant setting are eligible if > 6 months from treatment to metastatic diagnosis). 6. Has protocol-defined adequate organ and bone marrow function. 7. ECOG performance status 0 or 1.

4.2 Key Exclusion Criteria 1. Ineligible for any of the agents on the study. 2. Any substance abuse or other medical conditions that, in the investigator's opinion, may interfere with the subject's participation or study results. 3. Patients with spinal cord compression or clinically active central nervous system (CNS) metastases. Participants with clinically inactive brain metastases or treated brain metastases that are no longer symptomatic may be included in the study. 4. Active or prior documented (non-infectious) interstitial lung disease (ILD)/pneumonitis that required steroids, or suspected ILD/pneumonitis that cannot be ruled out by imaging at screening. 5. Prior randomization or treatment in a previous trastuzumab deruxtecan study regardless of treatment arm assignment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered via IV infusion every 3 weeks. Patients receive either: 1) T-DXd (5.4 mg/kg) + Pertuzumab (840 mg loading, 420 mg maintenance), 2) T-DXd (5.4 mg/kg) + Placebo, or 3) Standard of care THP regimen.
Route of Administration: Intravenous (IV) **Duration of Treatment:** Until unacceptable toxicity or objective disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptomatic management (e.g., antiemetics, pain control). A recognized and permitted pain management option includes **oxycodone** (Drug Preferred Name: oxycodone; ATC Code:

N02AA05). *Oxycodone Trade Names mapped for this protocol include:* Primlev, Oxecta, Xolox, Primalev, Endodan Reformulated May 2009, Percodan Reformulated May 2009, Magnacet, Dazidox, Endocodone, ETH-Oxydose, Perloxx, Narvox, Taxadone, Oxynorm, Oxyrapid, Combunox, Oxydose, Prolate, Tylox, Endocet, M-Oxy, Oxycontin, Oxyfast, Percolone, Roxicodone, Roxicet, Roxiprin, Roxilox, Percodan, Percocet, Endodan, Roxybond, Nalocet, Oxaydo, Xtampza, Troxyca, Xartemis, Lynox, Targiniq. **Prohibited:** Prior treatment with an antibody-drug conjugate (ADC) comprising a topoisomerase I inhibitor, or prior treatment with T-DXd/other anti-HER2 ADCs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, HER2 Status Confirmation, Echocardiogram/MUGA
Baseline	Day 0	Randomization, First IV Dose
Treatment Cycles	Every 3 Weeks	IV Infusion, Safety Labs, Efficacy Assessment, ILD Monitoring
End of Study	Variable	Disease Progression/Unacceptable Toxicity, Final Vital Signs, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Progression Free Survival (PFS)** by **Blinded Independent Central Review (BICR)** assessment: Defined as the time from the date of randomization until the date of objective radiological disease progression according to BICR using RECIST 1.1 or death by any cause.

7.2 Secondary Outcomes 1. **Progression Free Survival (PFS)** by Investigator assessment 2. **Overall Survival (OS)** 3. **Objective Response Rate (ORR)** by BICR and Investigator assessment 4. **Duration of Response (DoR)**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous assessment at every visit. Specific mandated monitoring protocols are in place for neutropenia, left ventricular dysfunction, and Interstitial Lung Disease (ILD)/pneumonitis. **Serious Adverse Events (SAEs):** Standard reporting timelines, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) established to review safety and efficacy signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses; Safety Set for all participants who received at least one dose of the study drug. **Sample Size Justification:** Target sample of $N=1,157$ is powered to test for the superiority of PFS in the T-DXd-containing arms versus the THP active control arm. **Handling of Missing Data:** Standard survival analysis methodologies, utilizing Kaplan-Meier estimation for time-to-event endpoints (PFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and safety. **Audit Trail:** eCRF locking, ongoing query resolutions, and independent central imaging review for tumor assessments.

11. Study Identifiers & Governance

Primary ID: D967UC00001 **Posting ID:** 272915588513506993 **Registry Number:** NCT04784715 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** DESTINY-Breast09

12. Clinical Framework & Disease Ontology

Disease Areas: Breast Cancer; HER2-positive; Metastatic **Condition Name:** Breast Cancer **Preferred Name:** Malignant neoplasm of breast **Preferred UMLS Name:** HER2-Positive Metastatic Breast Carcinoma **Semantic Type:** Neoplastic Process **Definition:** A primary or metastatic malignant neoplasm involving the breast. The vast majority of cases are carcinomas arising from the breast parenchyma or the nipple. Malignant breast neoplasms occur more frequently in females than in males. **MeSH Code:** D001943 **SNOMED ID:** 254837009 **SNOMED Hierarchy:** Malignant neoplasm of thorax | Malignant neoplastic disease | Neoplasm of thorax | Disorder of thorax | Disorder of thoracic segment of trunk | Disorder of trunk | Disease | Neoplasm of trunk | Neoplastic disease | Neoplasm of breast | Lesion of breast | Disorder of breast | Breast structure | Body organ structure | Anatomical structure | Anatomical or acquired body structure | Body structure | Structure of mammary region | Pectoral region structure | Structure of half of thorax lateral to midsagittal plane | Structure of half of trunk lateral to midsagittal plane | Structure of half of body lateral to midsagittal plane | Body region structure | Structure of subregion of trunk | Trunk structure | Body part structure | Thoracic structure | Chest and/

or abdomen structure | Chest and/or abdomen and/or pelvis structure | Neck and/or chest and/or abdomen structure | Structure of neck and/or chest and/or abdomen and/or pelvis | Neck and/or chest structure | Structure of thoracic segment of trunk | Structure of thoracic cross-sectional segment of trunk | Upper body part structure | Upper body structure | Malignant neoplasm. **Source Level:** 5

13. Study Procedures & Methodology

Management Pathway: 1st-line Metastatic Breast Cancer Treatment
Education Protocol (HCP): Site initiation and training focused on proactive ILD/pneumonitis monitoring and management **Education Protocol (Patient):** Comprehensive AE reporting education and ILD symptom awareness training **Quality Audit Mechanism:** Central imaging review (BICR) and global safety data monitoring

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity (approx. 60 months) **Onsite Visit Cadence:** Every 3 weeks for treatment cycles **Remote Monitoring Frequency:** As per site standard practices **Checklist Review Interval:** Tumor assessments every 6 to 9 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				